Nikko J. Cleri

Eberly Postdoctoral Fellow
The Pennsylvania State University

Summary

Research:	Galaxy Evolution, High-Redshift Galaxies, Emission-Line Galaxies, Interstellar Medium, Active Galactic Nuclei, Black Hole Seeds, Population III Stars, Star Formation, Dust Attenuation
Techniques:	UV/Optical/Near-IR Spectroscopy, Photoionization Modeling
Proposals:	As PI: 2 JWST, 2 Roman, 1 HST, 29.2 JWST hours and >0.8M USD awarded As Co-I: 14 JWST, 1 Roman, 1 Gemini, 1 HET, >850 total hours observing time
Publications:	9 as lead/co-lead author, 122 total, 9746 citations, 49 h-index
Presentations:	25 research, 18 outreach and professional development

Academic and Professional Appointments

Eberly Postdoctoral Fellow – The Pennsylvania State University	2024-present
Visiting Scientist – International Space Science Institute	2026
Graduate Student (Ph.D.) – Texas A&M University	2021-2024
Graduate Student (M.S.) – University of Connecticut	2019-2021
Research Assistant (B.S.) – University of Connecticut	2017-2019
NSF REU Student – Texas A&M University	2018

Education

Texas A&M University
Ph.D. in Astronomy — 2021-2024
Thesis: *Spectroscopic Studies of Stars and Black Holes Across Cosmic Time*
Advisor: Casey Papovich

University of Connecticut
M.S. in Physics — 2019-2021
Thesis: *CLEAR: Paschen-β Star Formation Rates and Dust Attenuation in Low Redshift Galaxies*
Advisor: Jonathan Trump
B.S. in Physics, Mathematics Minor — 2015-2019

Awarded Proposals and Grants

Principal Investigator — 5

2026	Roman 19082: *Characterizing the Interstellar Medium with Stacked Spectra of Ten Million Galaxies*	$150k
2026	Roman 19067: *A Census of Black Hole Evolution using High-Ionization Emission Lines*	$150k
2026	JWST-GO-10518: *CEDAR: Characterizing Electron Densities At Reionization*	29.2 hours/$234k
2024	JWST-AR-5558: *A Census of Optical Diagnostics of Ionizing Sources Across Cosmic Time*	$174k
2021	HST-AR-16609: *Peering Through the Dust: Paschen-beta Indicators of Star Formation and Dust Attenuation*	$136k

2026	Roman 19100: *A Shining Cosmic Dawn: Constraining Luminosity Functions and the Clustering of Luminous Galaxies at z > 8 with the Roman High-Latitude Wide-Area Survey* (PI: J. Helton)	$450k
2026	JWST-GO-11376: *Water in Little Red Dots? A Pathfinder Program with the MIRI/LRS to Confirm the Cool Temperatures of LRDs* (PIs: B. Wang, J. Helton)	13.0 hours
2026	JWST-GO-10445: *Reddened by dust or gas? Unveiling the "Red" in Little Red Dots with MIRI/LRS* (PIs: A. de Graaff, D. Setton)	49.1 hours
2026	JWST-GO-10311: *Slicing the Saguaro: Dissecting a Little Red Dot Descendant at Cosmic Noon with NIRSpec/IFU* (PIs: P. Rinaldi, E. Iani)	26.7 hours
2025	HET-PSU26-1-021: *The Hobby-Eberly Telescope Void Galaxy Sample* (PI: O. Curtis)	26 hours
2025	JWST-GO-8047: *Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking object by hunting for stellar absorption features* (PIs: B. Wang, E. Nelson)	19.6 hours
2025	JWST-GO-7488: *Echoes of Silence: Absorption Line Spectroscopy of a Massive Quiescent Galaxy at z=7.3* (PI: A. Weibel)	14.1 hours
2025	JWST-GO-8559: *SPAM: Star-formation from Photometry through the Addition of Medium-bands* (PIs: K. Davis, R. Larson)	62.8 hours
2025	JWST-GO-8410: *A Census of Galaxy Kinematics and Outflows to $z \sim 7$* (PI: R. Simons)	110.0 hours
2025	JWST-GO-8204: *Give me a break: the search for stars in a prototypical Little Red Dot* (PIs: J. Greene, I. Labbe)	17.4 hours
2024	JWST-GO-5718: *A Spectroscopic Census of Faint, Broad-Line AGN at z>5* (PIs: D. Kocevski, J. Guo)	20.49 hours
2024	JWST-GO-5943: *What really are the Physical Properties of Galaxies in the Epoch of Reionization?* (PIs: C. Papovich, W. Hu, T. Hutchison)	61.83 hours
2024	JWST-GO-5407: *MEOW: The MIRI Early Obscured-AGN Wide Survey* (PIs: G. Leung, R. Endsley, S. Finkelstein)	73.95 hours
2024	JWST-GO-5507: *Deep Spectroscopy of Galaxies at z=4-14: Uncovering Drivers of Early Galaxy Formation and Black Hole Growth* (PIs: T. Hutchison, R. Larson)	23.29 hours
2024	JWST-GO-6368: *The CANDELS-Area Prism Epoch of Reionization Survey (CAPERS)* (PI: M. Dickinson)	293.21 hours
2023	JWST-GO-3703: *Breaking the z=10 barrier with MIRI: redshift confirmation and detection of rest-frame optical emission lines* (PI: J. Zavala)	24.33 hours
2023	GS-2023A-Q-136: *Optical Spectroscopy of JWST ERO Galaxies* (PI: B. Backhaus)	20 hours

Mentoring

Graduate Students Mentored

Ieva Evaltaite, Penn State	2026-present
Owen Matthews Acuña, University of Wisconsin - paper submitted (OJAp)	2026-present
Quinn Casey, Dartmouth College - paper published (ApJ), paper submitted (ApJ)	2025-present

Undergraduate Students Mentored

Connor Luettgenau, Penn State	2025-present
Si Rui Ge, Nanjing University/Penn State ($\rightarrow$ graduate student at TDLI) - paper submitted (ApJ)	2025-present

Other Mentoring Activities

Co-Founder – Preparing the Emerging Next Generation Using Inclusive Networking (PENGUIN)

Mentor – Postdocs and Undergrads Linked for Summer Astrophysics Research (PULSAR) — Penn State
Mentor – Penn State-Nanjing Exchange Program — Penn State
Coordinator – Mentoring and Advising Graduates in an Inclusive Community (MAGIC) — Texas A&M
Mentor – UConn Undergraduate Peer Mentoring — UConn

Selected Honors and Awards

Dean's Climate and Diversity Award (Group) — 2024
Texas Space Grant Consortium Graduate Fellowship ($5K) — 2022
NSF Research Experiences for Undergraduates ($5K) — 2022
Governor's Scholarship ($8.5k/yr) — 2015-2019
Community Service Scholarship ($1k) — 2015

Teaching Experience

Instructor of Record
ASTRO 6: Stars, Galaxies, and the Universe — Penn State

Guest Lecturer
ASTRO 502: Fundamental Astrophysics — Penn State
ASTRO 480: Galaxies and Cosmology — Penn State

Teaching Assistant/Course Assistant
PHYS 1501: Physics for Engineers I — UConn
PHYS 1025: Introduction to Astronomy — UConn

Leadership and Service

Physics and Astronomy Community
JWST External Panelist, Galaxy Frontiers and the Distant Universe
Organizing Committee – 8th IGC Neighborhood Workshop, State College, PA, April 2026
Peer Reviewer – Astrophysical Journal (ApJ)
Peer Reviewer – Astronomy & Astrophysics (A&A)
Peer Reviewer – Monthly Notices of the Royal Astronomical Society (MNRAS)

College
Department Head Search Committee, Astronomy and Astrophysics — Penn State
Astronomy Representative – Eberly College of Science Postdoctoral Council — Penn State

Department
Committee Membership
Recruitment Committee, Outreach Committee, Climate and Diversity Committee — Penn State
Founder/Organizer – ExGal Journal Club — Penn State
Graduate Representative — Texas A&M
Organizer – Astronomy Journal Club — Texas A&M

Outreach

Presenter – Astronomy on Tap State College — Penn State
Volunteer – Gateway to Graduate School — Texas A&M
Demonstrator – Physics and Engineering Festival — Texas A&M
High School Research Reviewer – Lumiere — Texas A&M
Presenter – Astronomy on Tap BCS 'In the News' — Texas A&M
Treasurer – Astronomy on Tap BCS — Texas A&M
Pen-Pal – Letters to a Pre-Scientist — Texas A&M
Volunteer – Mitchell Institute Star Party Group — Texas A&M

Press

A Week Downeast, Challenging Assumptions about Galaxy Evolution, Astrobites, O. Cooper et al. 2025

Technical Skills and Programming Languages

Languages
Fluent: Python, LaTeX
Familiar: SQL, Julia, C, C++, R, IDL, perl, Mathematica, MATLAB, HTML, CSS

Software
Fluent: Cloudy, PyNeb, LiMe
Familiar: MAPPINGS, grizli, DS9, IRAF, sbatch, slurm

Website Architect

Personal Website: njcleri.github.io
Mentoring and Advising Graduates in an Inclusive Community (MAGIC) (co-author): tx.ag/tamumagic
Preparing the Emerging Next Generation Using Inclusive Networking (PENGUIN): penguin-mentoring.github.io

Publications

A current list of my papers can be found on SciX here.

Lead/Co-Lead/Student-Led Papers

† = Undergraduate-led papers

9. **Cleri, N. J.**, et al. 2026, *LEGGOS: A Shocking Lack of Evidence for Shocks at sub-kiloparsec Scales at 2 < z < 4*, arXiv e-prints, arXiv:2609.07850 (submitted to ApJ)
8. **Cleri, N. J.**, et al. 2026, *RUBIES: The Evolution of the Ionization Parameter from* $0 < z < 9$, The Open Journal of Astrophysics, 9, 71548
7. †Ge, S.-R., **Cleri, N. J.**, et al. 2025, *JWST/NIRSpec Reveals a Small Population of Dominant Dust-Obscured Ionizing Sources in Galaxies at 1 < z < 3*, arXiv e-prints, arXiv:2511.08671 (submitted to ApJ)
6. **Cleri, N. J.**, Olivier, G. M., Backhaus, B. E., et al. 2025, *Optical Strong Line Ratios Cannot Distinguish Between Stellar Populations and Accreting Black Holes at High Ionization Parameters and Low Metallicities*, ApJ, 994, 146,
5. Backhaus, B. E., **Cleri, N. J.**, et al. 2025, *Emission-Line Diagnostics at z>4: [O III]λ4364/Hγ versus [Ne III]λ3870/[O II]λ3730*, ApJ, 994, 125
4. **Cleri, N. J.**, Olivier, G. M., Hutchison T. A., et al. 2023, *Using [Ne V]/[Ne III] to Understand the Nature of Extreme-Ionization Galaxies*, ApJ, 953, 10
3. **Cleri, N. J.**, Yang, G., Papovich, C., et al. 2023, *CLEAR: High-Ionization [Ne V] λ3426 Emission-line Galaxies at* $1.4 < z < 2.3$, ApJ, 948, 112
2. **Cleri, N. J.**, Trump, J. R., Backhaus, B. E., et al. 2022, *CLEAR: Paschen-β Star Formation Rates and Dust Attenuation of Low Redshift Galaxies*, ApJ, 929, 3
1. **Cleri, N. J.**, Dunne, G. V., 2020, *Resurgent trans-series for generalized Hastings-McLeod solutions*, Journal of Physics A: Mathematical General, 53, 355203

Significant Author

8. Bolda, C. Papovich, C., Shen, L., **Cleri, N. J.**, et al. 2026, *NGDEEP: Anchoring the* $0.5 < z < 3.5$ *Star-Forming Main Sequence at Low Stellar Masses with Ultradeep JWST/NIRISS WFSS*, arXiv e-prints, arXiv:2610.40796

7. Helton, J. M., Witstok, J., Laska, M. Hainline, K. N., **Cleri, N. J.**, et al. 2026, *Discovery of the Lyman-α Forest in a Star-Forming Galaxy at $z \approx 7.57$: Implications for the Timing and Topology of Cosmic Reionization*, arXiv e-prints, arXiv:2609.38080

6. Matthews Acuña, O. S., Tremonti, C. A., **Cleri, N. J.**, et al. 2026, *The [Ne V] AGN Diagnostic in SDSS-IV/eBOSS: An Anticorrelation Between Ionization State and AGN Luminosity in Low-Redshift Coronal-Line Galaxies*, arXiv e-prints, arXiv:2608.21280

5. Casey, Q. O., Hickox, R. C., **Cleri, N. J.**, et al. 2026, *A Population of Little Red Dot-like Quasars in SDSS*, ApJ, 1009, 220

4. Larson, R. L., Finkelstein, S. L., Kocevski, D. D., Hutchison, T. A., Trump, J. R., Arrabal Haro, P., Bromm, V., **Cleri, N. J.**, et al. 2023, *A CEERS Discovery of an Accreting Supermassive Black Hole 570 Myr after the Big Bang: Identifying a Progenitor of Massive z > 6 Quasars*, ApJL, 953, L29

3. Backhaus, B. E., Bridge J. S., Trump, J. R., **Cleri, N. J.**, et al. 2023, *CLEAR: Detecting Low-Luminosity Active Galactic Nuclei at $0.6 < z < 1.3$ via Spatially Resolved Hubble Space Telescope Grism Emission Line Ratios*, ApJ, 943, 37

2. Prescott, M. K. M., Finlator, K. M., **Cleri, N. J.**, et al. 2022, *Using Multiple Emission Line Ratios to Constrain the Slope of the Dust Attenuation Law*, ApJ, 928, 71

1. Backhaus, B. E., Trump, J. R., **Cleri, N. J.**, et al. 2022, *CLEAR: Emission Line Ratios at Cosmic High Noon*, ApJ, 926, 161

Co-Author: Published

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77. Helton, J. M., et al. 2026, *Ionizing Photon Production Efficiencies and Chemical Abundances at Cosmic Dawn Revealed by Ultra-Deep Rest-Frame Optical Spectroscopy of JADES-GS-z14-0*, ApJL, 1004, L30

76. Brooks, M., et al. 2026, *Beyond the Monsters: A More Complete Census of Black Hole Activity at Cosmic Dawn*, ApJ, 1002, 129

75. Arevalo-Gonzalez, F., et al. 2026, *The AGN nature of strong CIII] emitters in the Early Universe with JWST*, A&A, 709, A46

74. Wang, B., et al. 2025, *The Missing Hard Photons of Little Red Dots: Their Incident Ionizing Spectra Resemble Massive Stars*, ApJ, 1003, 10

73. Llerena, M., et al. 2026, *Extreme equivalent width-selected low-mass starbursts at $z = 4 - 9$: insights into their role in cosmic reionization*, A&A, 708, A152

72. Papovich, C., et al. 2026, *Galaxies in the Epoch of Reionization Are All Bark and No Bite – Plenty of Ionizing Photons, Low Escape Fractions*, ApJ, 1000, 111

71. Perry, M. N., et al. 2025, *The Prevalence of Bursty Star Formation in Low-Mass Galaxies at z=1-7 from Hα-to-UV Diagnostics*, ApJ, 994, 14

70. Hviding, R. E., et al. 2026, *The X-Ray Dot: Exotic Dust or a Late-Stage Little Red Dot?*, ApJL, 1000, L18

69. Gandolfi, G., et al. 2026, *Mysteries of Capotauro - investigating the puzzling nature of an extreme F356W-dropout*, A&A, 706, A364,4

68. Zhang, Y., et al. 2026, *RUBIES Spectroscopically Confirms the High Number Density of Quiescent Galaxies from 2 < z < 5*, ApJ, 997, 252

67. Setton, D. J., et al. 2025, *Little Red Dots at an Inflection Point: Ubiquitous "V-Shaped" Turnover Consistently Occurs at the Balmer Limit*, ApJ, 995, 118

66. Hviding, R. E., et al. 2025, *RUBIES: A Spectroscopic Census of Little Red Dots; All V-Shaped Point Sources Have Broad Lines*, A&A, 702, A57

65. Mascia, S., et al. 2025, *Little impact of mergers and galaxy morphology on the production and escape of ionizing photons in the early Universe*, A&A, 701, A122,

64. Leung, G. C. K., et al. 2025, *Exploring the Nature of Little Red Dots: Constraints on AGN and Stellar Contributions from PRIMER MIRI Imaging*, ApJ, 992, 26

63. Setton, D. J., et al. 2025, *A Confirmed Deficit of Hot and Cold Dust Emission in the Most Luminous Little Red Dots*, ApJL, 991, L10

62. Katz, H., et al. 2025, *21 Balmer Jump Street: The Nebular Continuum at High Redshift and Implications for the Bright Galaxy Problem, UV Continuum Slopes, and Early Stellar Populations*, The Open Journal of Astrophysics, 8, 104

61. Kokorev, V., et al. 2025, *CAPERS Observations of Two UV-Bright Galaxies at z>10. More Evidence for Bursting Star Formation in the Early Universe*, ApJL, 988, L10

60. Burgarella, D., et al. 2025, *CEERS: Possibly forging the first dust grains in the universe: A population of galaxies with spectroscopically derived extremely low dust attenuation (GELDA) at 4.0 < z < 11.4*, A&A, 699, A336

59. Taylor, A. J., et al. 2025, *Broad-line AGNs at 3.5 < z < 6: The Black Hole Mass Function and a Connection with Little Red Dots*, ApJ, 986, 165

58. Llerena, M. et al. 2025, *The ionizing photon production efficiency of star-forming galaxies at* $z \sim 4-10$, A&A, 698, A302

57. Brooks, M., et al. 2025, *Here There Be (Dusty) Monsters: High-redshift Active Galactic Nuclei Are Dustier than Their Hosts*, ApJ, 986, 177

56. Kocevski, D. D., et al. 2025, *The Rise of Faint, Red Active Galactic Nuclei at z > 4: A Sample of Little Red Dots in the JWST Extragalactic Legacy Fields*, ApJ, 986, 126

55. Wang, B., et al. 2025, *RUBIES: JWST/NIRSpec Confirmation of an Infrared-luminous, Broad-line Little Red Dot with an Ionized Outflow*, ApJ, 984, 121

54. de Graaff, A., et al. 2025, *RUBIES: a complete census of the bright and red distant Universe with JWST/NIRSpec*, A&A, 697, A189,

53. Finkelstein, S. L., et al. 2025, *The Cosmic Evolution Early Release Science Survey (CEERS)*, ApJL, 983, L4

52. Weibel, A., et al. 2025, *RUBIES Reveals a Massive Quiescent Galaxy at z=7.3*, ApJ, 983, 11

51. Cooper, O. R., et al. 2025, *RUBIES: JWST/NIRSpec resolves evolutionary phases of dusty star-forming galaxies at* $z \sim 2$, ApJ, 982, 125

50. de Graaff, A., et al. 2025, *Efficient formation of a massive quiescent galaxy at redshift 4.9*, Nature Astronomy 9, 280

49. Shen, L., et al. 2025, *NGDEEP: The Star Formation and Ionization Properties of Galaxies at* $1.7 < z < 3.4$, ApJL, 980, L45

48. Cole, J. W., et al. 2025, *CEERS: Increasing Scatter along the Star-forming Main Sequence Indicates Early Galaxies Form in Bursts*, ApJ, 979, 193

47. Zavala, J., et al. 2025, *A luminous and young galaxy at z = 12.33 revealed by a JWST/MIRI detection of H*α *and [O III]*, Nature Astronomy, 9, 155

46. Cheng, Y., et al. 2025, *Unveiling the Dark Side of UV/Optical Bright Galaxies: Optically Thick Dust Absorption*, ApJ, 979, 71

45. Bisigello, L., et al. 2025, *Spectroscopic confirmation of a dust-obscured, metal-rich dwarf galaxy at z∼5*, A&A, 693, L18

44. Rose, C., et al. 2024, *CEERS Key Paper. IX. Identifying Galaxy Mergers in CEERS NIRCam Images Using Random Forests and Convolutional Neural Networks*, ApJL, 976, L8

43. Llerena, M., et al. 2024, *Physical properties of extreme emission-line galaxies at $z \sim 4-9$ from the JWST CEERS survey*, A&A, 691, A59

42. Gupta, A. R., et al. 2024, *Emission-Line Ratios and Ionization Conditions of CEERS Star-Forming Galaxies with JWST/NIRSpec*, Research Notes of the American Astronomical Society, 8, 266

41. Davis, K., et al. 2024, *A Census from JWST of Extreme Emission-line Galaxies Spanning the Epoch of Reionization in CEERS*, ApJ, 974, 42

40. Calabró, A., et al. 2024, *The evolution of the star formation rate and Σ_{SFR} of galaxies in cosmic morning ($4 < z < 10$)*, A&A, 690, A290

39. Chworowsky, K., et al. 2024, *Evidence for a Shallow Evolution in the Volume Densities of Massive Galaxies at $z = 4$ to 8 from CEERS*, AJ, 168, 113

38. Seillé, L.-M., et al. 2024, *Physical properties of strong 1 < z < 3 Balmer and Paschen lines emitters observed with JWST*, A&A, 689, A102

37. Hu, W., et al. 2024, *Characterizing the Average Interstellar Medium Conditions of Galaxies at $z \sim$ 5.6-9 with UV and Optical Nebular Lines*, ApJ, 971, 21

36. Napolitano, L., et al. 2024, *'Peering into cosmic reionization: the Lyα visibility evolution from galaxies at z = 4.5-8.5 with JWST*, A&A, 688, A106

35. Ronayne, K., et al. 2024, *CEERS: 7.7 μm PAH Star Formation Rate Calibration with JWST MIRI*, ApJ, 970, 61

34. Wang, B., et al. 2024, *RUBIES: Evolved Stellar Populations with Extended Formation Histories at $z \sim 7-8$ in Candidate Massive Galaxies Identified with JWST/NIRSpec*, ApJL, 969, L13

33. Finkelstein, S. L., et al. 2024, *The Complete CEERS Early Universe Galaxy Sample: A Surprisingly Slow Evolution of the Space Density of Bright Galaxies at $z \sim 8.5-14.5$*, ApJL, 969, L2

32. Pirzkal, N., et al. 2024, *The Next Generation Deep Extragalactic Exploratory Public Near-Infrared Slitless Survey Epoch 1 (NGDEEP-NISS1): Extra-Galactic Star-formation and Active Galactic Nuclei at 0.5 < z < 3.6*, ApJ, 969, 90

31. Jung, I., et al. 2024, *CEERS: Diversity of Lyman-Alpha Emitters during the Epoch of Reionization*, ApJ, 967, 73

30. Mascia, S. et al. 2024 *New insight on the nature of cosmic reionizers from the CEERS survey* A&A, 685, A3

29. Morales, A. M., et al. 2024, *Rest-Frame UV Colors for Faint Galaxies at $z \sim 9-16$ with the JWST NGDEEP Survey*, ApJL, 964, L24

28. Cheng, Y., et al. 2024, *Exploring the Gas-Phase Metallicity Gradients of Star-forming Galaxies at Cosmic Noon*, ApJ, 964, 94

27. Shen, L., et al. 2024, *NGDEEP Epoch 1: Spatially Resolved Hα Observations of Disk and Bulge Growth in Star-Forming Galaxies at $z \sim$ 0.6-2.2 from JWST NIRISS Slitless Spectroscopy*, ApJL, 963, L49

26. Barro, G., et al. 2024, *Extremely Red Galaxies at z = 5–9 with MIRI and NIRSpec: Dusty Galaxies or Obscured Active Galactic Nuclei?*, ApJ, 963, 128

25. Backhaus, B. E., et al. 2024, *CEERS Key Paper. VIII. Emission-line Ratios from NIRSpec and NIRCam Wide-Field Slitless Spectroscopy at z > 2*, ApJ, 962, 195

24. Kirkpatrick, A., et al. 2023, *CEERS Key Paper VII: JWST/MIRI Reveals a Faint Population of Galaxies at Cosmic Noon Unseen by Spitzer*, ApJL, 959, L7

23. Calabró, A., et al. 2023, *Near-infrared emission line diagnostics for AGN from the local Universe to redshift 3*, A&A, 679, A80

22. Fujimoto, S., et al. 2023, *ALMA FIR View of Ultra High-redshift Galaxy Candidates at $z \sim$ 11-17: Blue Monsters or Low-z Red Interlopers?*, ApJ, 955, 130

21. Kocevski, D. D., et al. 2023, *Hidden Little Monsters: Spectroscopic Identification of Low-Mass, Broad-Line AGN at $z > 5$ with CEERS*, ApJL, 954, L4
20. Arrabal Haro, P., et al. 2023, *Spectroscopic confirmation of CEERS NIRCam-selected galaxies at $z \simeq 8 - 10$*, ApJL, 951, L22
19. Estrada-Carpenter, V., et al. 2023, *CLEAR: The Morphological Evolution of Galaxies in the Green Valley*, ApJ, 951, 115
18. Yang, G., et al. 2023, *CEERS Key Paper VI: JWST/MIRI Uncovers a Large Population of Obscured AGN at High Redshifts*, ApJL, 950, L5
17. Papovich, C., et al. 2023, *CEERS Key Paper IV: Galaxies at $4 < z < 9$ are Bluer than They Appear – Characterizing Galaxy Stellar Populations from Rest-Frame ~ 1 micron Imaging*, ApJL, 949, L18
16. Simons, R. C., et al. 2023, *CLEAR: Survey Overview, Data Analysis and Products*, ApJS, 266, 13
15. Constantin, L. et al. 2023, *Expectations of the size evolution of massive galaxies at $3 \leq z \leq 6$ from the TNG50 simulation: the CEERS/JWST view*, ApJ, 946, 71
14. Perez-Gonzalez, P. G., et al. 2023, *CEERS Key Paper V: A triality on the nature of HST-dark galaxies*, ApJL, 946, L16
13. Kocevski, D. D., et al. 2023, *CEERS Key Paper II: The Resolved Host Properties of AGN at 3 < z < 5 with JWST*, ApJL, 946, L14
12. Finkelstein, S. L., et al. 2023, *CEERS Key Paper I: An Early Look into the First 500 Myr of Galaxy Formation with JWST*, ApJL, 946, L13
11. Guo, Y. et al. 2023, *First Look at z > 1 Bars in the Rest-Frame Near-Infrared with JWST Early CEERS Imaging*, ApJL, 945, L10
10. Trump, J. R. et al. 2023, *The Physical Conditions of Emission-Line Galaxies at Cosmic Dawn from JWST/NIRSpec Spectroscopy in the SMACS 0723 Early Release Observations*, ApJ, 945, 35
9. García-Argumánez, A. et al. 2023, *Probing the earliest phases in the formation of massive galaxies with simulated HST+JWST imaging data from Illustris*, ApJ, 944, 3
8. Zavala, J. et al. 2023, *Dusty starbursts masquerading as ultra high redshift galaxies in JWST observations*, ApJL, 943, L9
7. Rose, C. et al. 2023, *Identifying Galaxy Mergers in Simulated CEERS NIRCam Images using Random Forests*, ApJ, 942, 54
6. Finkelstein, S. L. et al. 2022, *A Long Time Ago in a Galaxy Far, Far Away: A Candidate $z \sim 14$ Galaxy in Early JWST CEERS Imaging*, ApJL, 940, L55
5. Papovich, C. et al. 2022, *CLEAR: The Ionization and Chemical-Enrichment Properties of Galaxies at $1.1 < z < 2.3$* ApJ, 937, 22
4. Matharu, J. et al. 2022, *CLEAR: The Evolution of Spatially Resolved Star Formation in Galaxies between $0.5 \leq z \leq 1.7$ using Hα Emission Line Maps*, ApJ, 937, 16
3. Jung, I. et al. 2022, *CLEAR: Boosted Lyα Transmission of the Intergalactic Medium in UV bright Galaxies*, ApJ, 933, 87
2. Simons, R. C. et al. 2021, *CLEAR: The Gas-Phase Metallicity Gradients of Star-Forming Galaxies at $0.6 < z < 2.6$*, ApJ, 923, 203
1. Estrada-Carpenter, V. et al. 2020, *CLEAR II: Evidence for Early Formation of the Most Compact Quiescent Galaxies at High Redshift*, ApJ, 880, 2

Co-Author: Submitted

28. Larson, R. L., et al. 2026, *SPAM: Survey Overview and JWST/NIRCam Multi-Band Photometry Catalog*, arXiv e-prints, arXiv:2610.35570

27. Shen, L., et al. 2026, *Quiescent Galaxies at $z > 2$ from CAPERS spectroscopy and their Number densities*, arXiv e-prints, arXiv:2609.20931

26. Wang, B., et al. 2026, *Balmer Absorption Series and Broad Metal Lines in Two Luminous Little Red Dots*, arXiv e-prints, arXiv:2609.22623

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Presentations

Research Presentations (**bold** = invited)

25.	**Talk**: *Interpreting the Rest-Frame Optical Spectra of High-Redshift Galaxies*, at the Carnegie Observatories, Pasadena, California, USA	9 October 2026
24.	Talk: *Interpreting the Rest-Frame Optical Spectra of High-Redshift Galaxies*, at the University of California, Los Angeles, Los Angeles, California, USA	8 October 2026
23.	**Talk**: *The Rest-Frame Optical Spectra of High-Redshift Galaxies: What are we seeing?*, at the California Institute of Technology, virtual	8 June 2026
22.	**Talk**: *The Rest-Frame Optical Spectra of High-Redshift Galaxies: What are we seeing?*, at the Dartmouth College, Hanover, New Hampshire, USA	1 May 2026
21.	**Talk**: *Characterizing High-Redshift Sources with Optical Emission Line Ratios*, at the Space Telescope Science Institute, Baltimore, Maryland, USA	14 August 2025
20.	Talk: *Characterizing High-Redshift Objects with Optical Strong Line Ratios*, Winter Harbor, Maine, USA	28 July 2025
19.	Talk: *A Census of Optical Diagnostics of Ionizing Sources Across Cosmic Time*, Bergen, Netherlands	8 May 2025
18.	Talk: *High-Redshift Diagnostics of Star Formation and Active Galactic Nuclei*, The Pennsylvania State University, State College, Pennsylvania, USA	10 February 2025
17.	Talk: *High-Redshift Diagnostics of Star Formation and Active Galactic Nuclei*, The Pennsylvania State University, State College, Pennsylvania, USA	17 September 2024
16.	Talk: *High-Redshift Diagnostics of Ionizing Sources*, University of Wisconsin, Madison, Wisconsin, USA	17 July 2024
15.	Discussion Chair: *Active Galactic Nuclei and Little Red Dots*, San Lorenzo de El Escorial, Spain	13 May 2024
14.	Talk: *Diagnostics of Ionizing Sources at High-z using Models and Observations*, San Lorenzo de El Escorial, Spain	13 May 2024

13.	Talk: *Diagnostics of AGN, Black Hole Seeds, and Population III Stars with JWST* at the AAS 243rd Meeting, New Orleans, Louisiana, USA	10 January 2024
12.	Poster: *Emission Line Ratio Diagnostics of AGN, Black Hole Seeds and Population III Stars with JWST* at the First Year of JWST Science Conference, Space Telescope Science Institute, Baltimore, Maryland, USA	11 September 2023
11.	Talk: *Diagnostics of Exotic Ionizing Sources with JWST* at Texas A&M Astrosymposium, College Station, Texas, USA	17 August 2023
10.	Talk: *Diagnostics of Exotic Ionizing Sources Across Cosmic Time - High-Ionization Emission-Line Ratios: Ne53* at University of Texas, Austin, Texas, USA	10 May 2023
9.	Poster: *High-Ionization [Ne V] Emission-Line Galaxies at Cosmic Noon and the Epoch of Reionization* at AAS 241st Meeting, Seattle, Washington, USA	12 January 2023
8.	Talk: *Using [Ne V] to Constrain the Sources of Highly-Energetic Photoionization Across Cosmic Time: Exploring the "Mystery of Neon" with HST and JWST* at Texas A&M University, College Station, Texas, USA	2 December 2022
7.	Talk: *Extreme High-Ionization Emission-Line Galaxies at Cosmic Noon and the Epoch of Reionization: Exploring the "Mystery of Neon" with HST and JWST* at Texas A&M University, College Station, Texas, USA	18 August 2022
6.	Talk: *The Evolution of Spectroscopy from HST to JWST: Implications for the Epoch of Reionization* at Texas A&M University, College Station, Texas, USA	22 July 2022
5.	Poster: *HST Grism Observations of Paschen-Line Star-Formation and Dust Attenuation: A Precursor to the JWST Era* at AAS 240th Meeting, Pasadena, California, USA	14 June 2022
4.	Talk: *Paschen-β Star Formation Rates and Dust Attenuation with HST and JWST* at Texas A&M Astrosymposium, College Station, Texas, USA	27 August 2021
3.	Poster: *CLEAR: Paschen-β Star Formation Rates and Dust Attenuation in Low Redshift Galaxies* at AAS 237th Meeting, Virtual	13 January 2021
2.	Poster: *Modeling ^{8}B Solar Neutrino Detection with CEνNS* at AAS 233rd Meeting, Seattle, Washington, USA	9 January 2019
1.	Poster: *Modeling ^{8}B Solar Neutrino Detection with CEνNS* at TAMU Undergraduate Research Poster Session, College Station, Texas, USA	1 August 2018

Outreach Presentations

7.	"Broken Lecture": *Did JWST "Break The Universe"?*, State College, Pennsylvania, USA	29 April 2026
6.	Talk: *Did JWST "Break The Universe"?*, State College, Pennsylvania, USA	11 April 2026
5.	Talk: *Island Universes: Is the Milky Way Everything?* at Astronomy on Tap State College, State College, Pennsylvania, USA	19 March 2026
4.	Talk: *Astronomy and You: The Impacts of Astronomy on Everyday Life* at Astronomy on Tap State College, State College, Pennsylvania, USA	20 March 2025
3.	Talk: *The Evolving Universe Through JWST's Eyes* at the Pennsylvania State University, State College, Pennsylvania	1 February 2025
2.	Talk: *The Origin of the Elements: Chemistry Across Cosmic Time* at Astronomy on Tap State College, State College, Pennsylvania, USA	19 September 2024

1.	Talk: *Beyond the Telescope: Unraveling Mysteries with AI in Astronomy* at Astronomy on Tap B/CS, Bryan, Texas, USA	24 April 2024

Professional Development Presentations

11.	Talk: *Life After Submission: Refereeing, Proofs Editing, And All The Other Boring Stuff That Nobody Prepares You For* at The Pennsylvania State University, State College, Pennsylvania, USA	10 June 2026
10.	Panel: *Applying to Postdocs* at The Pennsylvania State University, State College, Pennsylvania, USA	7 April 2026
9.	Panel: *Applying to Postdocs* at The Pennsylvania State University, State College, Pennsylvania, USA	24 March 2025
8.	Panel: *GLASS Postdoc Panel* at Texas A&M University, College Station, Texas, USA	8 March 2024
7.	Talk: *How to Be A Referee* at Texas A&M University, College Station, Texas, USA	10 November 2023
6.	Panel: *How to Get Into Grad School* at Texas A&M University, College Station, Texas, USA	28 July 2023
5.	Talk: *Data Visualization in Astronomy: More Important than the Science Itself?* at Texas A&M University, College Station, Texas, USA	11 November 2022
4.	Panel: *How to Get Into Grad School* at Texas A&M University, College Station, Texas, USA	29 July 2022
3.	*Data Visualization in Astronomy: More Important than the Science Itself?* at Texas A&M University, College Station, Texas, USA	2 June 2022
2.	Workshop: *Matplotlib: The Champion of Plotting in Python* at Texas A&M University, College Station, Texas, USA	2 June 2022
1.	Workshop: *pandas: Your Best Friend for Data Analysis in Python* at Texas A&M University, College Station, Texas, USA	1 June 2022

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